

## Implementing Responsible AI

*By Ashley Casovan, Executive Director, Responsible AI Institute*

*This module introduces the audience to the concept of responsible AI and the many factors that play a role in the design, development, deployment and use of responsible AI systems. It outlines key challenges and explains how different stakeholders may take different approaches in their mitigation strategies. This module also looks at the work the Responsible AI Institute (RAII) is doing in the field and on the creation of their formal certification system for responsible AI systems.*

### Lecture Transcript

**0:03** Hello, I'm Ashley Casovan, the Executive Director of the Responsible AI Institute, a nonprofit organization dedicated to the responsible use and oversight of AI systems. Today, I'm going to talk about how to implement responsible AI in a tangible and practical way, both for those that are designing, developing, deploying, and even using AI systems, and really happy and excited to be part of this Global AI ethics Consortium, and the great work that everybody sharing throughout these modules.

**0:41** So throughout what you've heard already, you've probably come to understand the impact that AI will have on our society, and how it's going to impact some people's lives differently than others. And at the Responsible AI Institute, we really believe in the promise of new technologies, like AI systems, and how they can really help us to solve some of the biggest global challenges that are facing us. But also just want to make sure that there's appropriate checks and balances, that they're built in a way that is trustworthy, transparent, fair. And from what you'll hear in terms of the approach that we take, we really care about that audibility or accessibility aspect of these technologies to make sure that ethics, human rights and broader responsibility goals are included as part of the review and implementation of these systems.

**1:47** And so I'm just putting up some quotes here on the screen, I won't go through all of them. But the idea really is to get a sense of not just the impact or influence that these technologies will have, but really how integrated into many different lines of business they are. We're seeing that maybe, you're thinking, when you think of AI, you think of things like Siri or Google Maps, things that are really in your face, or you understand you can see that, there's something that's artificial or augmenting human processes. But in many circumstances, there's AI technologies that are underlying a lot of the different services that are being provided. And in many ways we don't actually see. Within the Responsible AI Institute, we're focused primarily on thinking about the implications of AI from health, finance, human resources, and as well as thinking about the procurement of these technologies. But that just is some of the uses and some of the industries that they're being used in.

**3:02** And so for us, as you can see, kind of where I'm going at with some of my comments, and this is that, while again, we not only understand the impact, and really do support the use of these systems, we really want to make

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sure that there are appropriate guardrails that are put in place. Because what we've seen from the use of these technologies thus far is that sometimes they can hurt people in many different types of ways. But it could be things through the use of facial recognition technologies that are incorrectly identifying somebody and improperly imprisoning them as a result of that improper identification. And what we're seeing is given how these AI systems are built, when done so in a way that is being using historical data that might have some biases in it, or really just using data that's not complete or fit for purpose, then there's a lot of biases for particular demographics. And with facial recognition systems, we've seen this be particularly skewed negatively towards people of color. And it's easy to think about in that kind of matching patterns way when thinking about image and object recognition with technologies like facial recognition, but we can see these same types of biases happen with things like jail sentences, recidivism rates, for predicting when people's jail sentence should be extended, or or otherwise, but also things with access to credit that really enables us to have the means in order to advance our lives and the livelihood of those around us.

**4:57** And so we might think that just because one person gets more access to credit than the other, that it's not that big of a problem. But it's really the tip of the spear in terms of again, that access to wealth that can lead to greater livelihood. And so it's really important to think about these potential harms that can come from the use of these systems. We think about them in terms of intended consequences, or intentional, so that there's known trade-offs that are being taken into account.

**5:35** And then there's unintended consequences. Those are things that were just not potentially understood, when thinking about the implications or the development of the system, they just weren't taken into account. And then there's systemic harms that can come from the use of these systems. And those are things more so like, what we're seeing when there's automation in a loan application process is that based on a person's past experiences, there might be reasons why they are not as credit worthy as others but that could be from things like the color of their skin, or their gender, or their age. And those are not always great predictors of future behavior. But that's a main way in which these AI systems are really trying to learn and replicate human behavior. And so when there's a human in the process, there can be those intervention points that take into account these different variables. But when it's being done in an automated way, then we can see that in addition to even being hurtful, it can be exclusionary, or what we see in many of these circumstances that in order to understand behavior, there's a lot of monitoring that's going on. And so there can be infringement on people's privacy, or access to including, over access, to information.

**7:16** And so one thing to really take into account with this is that when thinking about the application of these systems, they're often happening, not only in different industries like I was mentioning before, but they're happening in different regions of the world. And as a result, there's different types of cultural norms and acceptances of these technologies. We've seen in China that there's a greater acceptance of things like facial recognition technologies, to try and have increased safety, or comfort within society. But we've also seen where there's been the use of these systems in things like testing, and as it relates to grading a bunch of grading exams. And we've seen this recently with SAT scores in the US and with A-Levels in the UK, how, given the the demographic, a younger demographic that understands technology, and the potential implications of it, understood that when the grades were skewed in a way that were unexpected, especially impacting those in lower income demographics in the case of the A-Levels, there was an understanding that that was something that was how that

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system was being built in a way that was jeopardizing those particular people. And so as a result, we can see that there's then action that's taken by, in this case those students, to try and reverse some of those decisions and create an appropriate recourse related to that. And so through these different mechanisms, we're really as policymakers trying to understand not only the implications of what can potentially happen, but also what we can do about it. And so just a few examples here to kind of articulate then some of those oversight mechanisms that I'll talk about in a little bit are. I wanted to share one in health, one in the labor context, and then one in the financial context. And one use case that we're working on right now with researchers at Memorial Sloan Kettering as well as an organization called Skinopathy is looking at the detection of skin ailments so skin cancer, burns, etc. And how a an automated process taking a picture with your phone and seeing a different lesion or burn that you might have, and that, maybe not giving you a diagnosis, but allowing you to have greater access to information, or triaging you so that you can get access to care faster.

**10:18** And so these are things though, that if the system were to give a diagnosis, which we're seeing many that do do that, and you either have even a false positive or a false negative, it really could lead to unnecessary harm. And so again, trying to balance these out, not saying not to have these systems, but how do you do so in a way that's responsible in protecting people. Similarly, within there's many of you are probably aware, especially given that you're in New York City, maybe not everybody. But the work that New York City is doing related to thinking about automated employment systems, and the implications that they could have. And one of the use cases that we're working on with the US Department of Labor is a recommendation system that is putting recently retired military vets into civilian roles and doing some job matching related to that. And we're really trying to understand the implications that could exist from that. So what you don't want to do is just because you drove a tank in the military, I then have a prediction system that says, okay, well, you can only work in transport types of jobs or roles in the future. There's probably lots of other great skill sets that one acquires as well in the military that might have, in this case, how these types of systems work is based on vocabularies. And so there'll be certain types of job descriptions, or job titles, that then can equate to other types of job descriptions or job titles. And we want to make sure that those jobs are then being coded in a way that it's clear what all of the various different skills that one acquires in those roles, is capable of doing and therefore have many options available to them in a civilian type of role.

**12:36** And so then that leads to one that I was mentioning earlier, just eligibility for access to credit. And again, allowing there to be these automated systems allows any of us that have borrowed money, so probably many of us are most of us on this, listening to this knows, it's really great when you can get that information back fairly seamlessly and quickly, as opposed to having a huge backlog that would take you months and months and months to get something like a credit card or a mortgage, or a loan for a car. And so the benefit of having these systems really does expedite these efforts. But it can be something that really can cause some challenges, when, as I was mentioning earlier, done disproportionately to different demographics. And so we've seen this with things like the Apple credit card, and in the same households giving a male one greater access to credit than the female.

**13:53** And so all of this has really led to a lower public trust in AI. This survey that I'm showing on the left hand side is from 2018. But I really like the particular question that Ipsos asked in this survey. And from what we've seen, the numbers really have not changed too too significantly. What I really liked about this one question was, the more that it matters, the lower the support for AI that exists. And I think that this is an interesting concept

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because as we were talking about with some of the previous examples, just not having these systems is not an option anymore. We've really come to depend on them. It's helping our society advance quicker, allow people to get increased access to credit, do things in a more, in a way that's more accurate, often, but without those guardrails around it, there's that diminishing trust within this.

**15:00** And that, as I said, while the study was done in 2018, newer work that KPMG did with University of Queensland in March 2021, reflects some of those similar concerns. I would definitely encourage you to check out both of these studies. But there's a lot of public sentiment that happens, types of evaluations that are happening regionally. And one thing that I've always found really interesting is the trust level differences between different demographics. So women versus men, younger people versus older people, etc. And, seeing those between different regions is something I would definitely encourage you to check out for your region if a study like that is available.

**15:48** And again, those are things just to review all of this are really things to keep in mind, if you're at all involved in the design, development, deployment, or use of AI systems. And so that's really the work that my organization Responsible AI Institute, or RAI as we refer to it, is working on in terms of thinking about, well, we have these opportunities with these various different systems: What is it that we can do, to make sure that they're functioning in ways that our hearts are helping people and, and helping society versus causing potential harms, especially when they can be avoided. And so that's why, if you read any of the literature related to this, there's often discussions about unintentional consequences or unintentional harms. And those are the ones that we really want to make sure at a minimum we're avoiding. And so the efforts that you'll see that we're working on are really working to increase that public trust, to make sure that we're taking advantage of those benefits from AI systems, and ensuring that then there's the appropriate protections that are in place for these different types of AI systems. And with that, recognizing that not all AI systems have the same degree of risk. So they might not need the same degree of oversight, but also that there's different harms associated with each of those. And so through that, really, overall want to reduce the negative impacts and incidents caused by AI systems, as well as accelerating positive business and societal impact. And some of you may be thinking why we're concerned about positive business impact.

**17:40** As we can see from programs like ESG, there's really opportunities to help drive business objectives and behaviors. And through that, we can set some of these, we can move some of these norms and desires for how we want to build these systems forward in a way that aligns with corporate goals and objectives as well. And so through that, there's been really understanding what it means to address some of these challenges and concerns has been a significant thought on both international organizations' minds, corporations' minds, governments, and then companies that are developing these systems. And so there's been a plethora of work that's been done, documentations that have been written to try and address these different issues through whether they be policies, whether they be tools that are really looking at certain aspects of bias, explainability, etc., to address some of those challenges, the challenges that are outlined. And through this, there's really in the period of 2017, 2016 to 2020. A lot of those documents resulted in high level principles. And they were adopted at international organizations like the OECD, UNESCO, World Economic Forum, etc. And through that, there's been a really great understanding of not just principles, but different types of codes of conduct that can be put in place, both within governments and

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companies and academic institutions to help guide some of those behavioral changes that I was mentioning before.

**19:46** One of the challenges though, and my background is coming into this from being the creator of one of those. I was the Director of Data and Digital for the Government of Canada, where I led the development of the directive on automated decision making systems, and through that we outlined a policy that had nine requirements in it that spoke to some of those principles. So I've mentioned bias and fairness, explainability and interpretability of these systems, but things that we're seeing, even in the New York City Law that I mentioned, related to automated employment systems are requiring things like notification, so notifying the user that an automated system is being used. So that in the case where there's an issue, that there could be appropriate recourse mechanisms that are identified and then made available to the people who are the subject of the use of that system.

**20:51** And so through that really what I've been working on with RAII is trying to understand that there's going to be these various different regulations that are going to come out from national, state, regional, local levels, as well as corporations are going to put their own type of laws, right not laws, they cannot put laws in, but rules in place to try and meet some of those objectives really coming from those principles and that could really be in line with their corporate values, and also normative or forming normative ESG objectives, which of many corporate boards are really concerned about and and tracking against. And so then the question becomes, how do you take those and then apply things that are existing that are currently discussed, I'm sorry, existing standards, as well as discussions that are underway for creating future standards that are helping to address some of those. Both compliance against the regulations, but address some of the concerns that are raised in these principles. And through that, we see the work of accreditations, management standards, and then also use case and function specific types of certifications. And that's really, the work that we're doing at RAII is looking at how can we take various different what I think of as task specific standards. So if we think about it in terms of a building, because it's just a little bit more tangible, things like your, the luminosity of a light ball, so that certain environmental objectives are being met, or things like your wall plug, so that it's standard between whether or not you're plugging in your computer or your your lights, etc. And so these are things that are very task specific or object specific.

**23:07** But you need many of those things when you're thinking about an AI system that contains many different components. And so through that, the work we're doing at RAII is to bring that those various different tasks specific standards for a particular type of system, which I'll talk about in a second, and really map that to some of the emerging management system standards that are being discussed in regulations like the draft EU AI policy, that are coming from international organizations like ISO. And so through that, one thing that you saw in one of the earlier quotes from Gartner, when at the start of the discussion, was around then the auditability. And so this is a really common way to think about accountability or oversight of systems is to not only have these types of different policies, and then standards that are put in place that are commonly agreed upon by governments, by industry, by individuals. But then there's a check and balance in terms of the evaluation or the audit of those practices. And so we're seeing that, that many organizations that currently perform audits are capable of doing those functions, which really leads me to also what we're seeing an absence of any of those principles or policies and regulations being enacted, or in full force yet, is really what corporate responsibility or what's happening at a corporate responsibility level, which is similar with ESG.

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**24:59** So environmental, social, and governance objectives, how those can influence, along with model risk management and then internal audit practices as well as systems operations practices that we see with things like enterprise architecture, review boards, and technology review practices, how all of those together can really work to influence the desired outcomes of a responsible AI review process. And through that, really ensuring that there's harmony between the operational side of the house or sometimes referred to, especially in the financial world, as one as the first kind of domain, and then or Line of Defense 1. And then the second domain being Line of Defense 2, being more of their risk management, oversight, legal practices or often internal legal, is within that LOD 2. And then the internal audit portion, really trying to make sure that there's appropriate checks and balances and oversight mechanisms, being that Third Line of Defense.

**26:18** And so these, at an organizational level, these different lines of defense or business units, are really looking at what's coming down the pipe in terms of future regulations, as well as different types of current or emerging standards, and really figuring out how they fit into their current organizational oversight practices. And so as I mentioned, when we're thinking about this in relation to an AI system, it's important to know what an AI system actually is. And so what are we talking about having oversight over? And so, as you can tell from the previous slides, some of these norms or standards or regulations might be required for a particular function like independent review, or a governance board being put in place, which would be the responsibility at a corporate level. But a corporation might have many, many different types of systems. And some might be really great and human rights protecting, and others might have some greater concerns. And like I was mentioning earlier, there could be different risk levels associated with the deployment of all of those. And so we really think that they there needs to be oversight at that system level, not just at a corporate level. And so this is the definition that we use of a system, but also in line with some of the efforts that are coming out of OECD in terms of really defining what the function of AI systems are. And so, it has a system has data in it, so that's feeding the model. And when I was exactly when I was speaking earlier about some of the different examples. And I talked about historical data, or different types of, the skin color or the potential issues with that, it could actually be a lack of data. So that I when the model is being trained, or learning, which I'll come to in a second, that it's only being fed data that's coming from people have one skin color. And so then it's really difficult for that system to recognize similar types of patterns with people of different types of skin color. So these are things to really take into consideration when thinking about the data that is representative, that is fit for purpose, that it is also being corrected for other types of historical biases that might exist.

**29:07** Similarly with models. Models are really like a recipe, kind of the instructions on how to do something. And so if those instructions aren't accurate, or they've been built with somebody's different, developer's perspective, with certain types of perspectives, and there aren't checks and balances on those, or they're not in line with some of the business objectives, different types of concerns like bias can creep into, into that model. Similarly, I've mentioned explainability a few times. In many circumstances, some of the greatest concern around these models is that it's not really clear how those models are working, what the recipe actually is. And especially in the case of machine learning, the data evolves and changes over time. And so is there, and that's why when we were talking about explainability, interpretability is also an option as well as, can you tell post facto or after how that model was functioning and operating. And then last but not least, is the data and the model can operate in a vacuum. But that's often not the case. Maybe that can be controlled when an AI system is being tested in a



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research way. But in the real world, with, there's always some sort of context in where it's being deployed, that could be the same type of model or system, could be used in different domains or industries, a prediction system could be even just using the examples we've already talked about, it could be predicting somebody's access to credit. And it could also be some of these could be predicting somebody's access to health care. And so but it could be the same type of model or system being used.

**31:03** And those have different types of implications. Similarly, with AI tasks, or the system type. Even in this, I've given examples of different types of AI systems, we've talked a lot about prediction models, which are the definitely the most common. But we also are seeing this in relation to image and object recognition, or things like chatbots. And so there's that computational interaction there. And so all of those different types of things, when people talk about AI, they could mean any of those, and also some of those together in an ensemble way within a model. And so that then has different implications in terms of how they work together. And then and also then the different types of oversight that's needed. And then finally, we have the geographic region. And so as I was mentioning earlier, some of the potential cultural differences between these different geographies really need to be taken into account.

**32:05** And so the work that we've done at RAIL is to really understand a lot of what I've talked about is very overwhelming, or it can be very overwhelming. It's really difficult sometimes to figure out who's doing what, in terms of policy development. What is it that a team that's designing these systems or deploying them really needs to be thinking about doing. And so I've been a practitioner, I come from a data architecture background. And for me, it's really important that we take the guesswork out of policy, that we don't create additional requirements for interpreting these big higher order principles, and really figure out what then that means, for then the design and development of AI systems. And so we've really tried to take that guesswork out of this and have an easy survey that or assessment that those that are building these systems can use to ensure that they're aligned with those objectives. And so we've worked, or we continue to work, I sit as a subject matter expert on OECD's AI Policy Observatory and so we've aligned with those globally adopted AI principles. However, you'll see that there's a lot of commonalities if you look at some of these other principles, amongst all of them.

**33:37** And so within that, we are really working to ensure that the implementation framework itself takes into account all of those different practices. And you'll see on the next slide that we've detailed then requirements that come from that, and that you'll see, as I move forward, is then evaluated through an assessment, as I was mentioning. And we've been building this work in the open with the public. And so I speed that be subject matter experts from industry, civil society, policymakers from government, and academia, that are really helping us to take best in class research, and integrate that into our understanding. But as I said, coming from being a practitioner, it's really been important to me that this is testing against real use cases. I think we can set objectives, but it's really difficult for them to actually be met. If we're not assessing them against various different use cases and understanding what trade offs do need to be made. And what's really important when we're tracking are really looking at the oversight of these systems. And ultimately, then we work to feed that back into standards development processes. So that that can help to inform that normative development process.

**35:02** And as I mentioned here, then taking that inner circle and really thinking about its implications, not just at that top level, but what types of requirements are needed in a more granular level. Our assessment, you have

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probably heard of impact assessments, it's really similar to that concept is we're evaluating both the risk and also the mitigation measures to get a residual risk score from that, recognizing that a system could have inherent risk. But if there's been good practices that have been put into account, that that's something that can really reduce the potential harm that could come from that system.

**35:49** And so as I was mentioning, this work really is done through an evaluation or that assessment. And so you can see, again, the breakdown of what we're talking about related to or what the scope of what we're assessing is. So related to that AI system being the data, the model and the context, taking into account, the system type, the geographic region and the domain or industry norms or rules that might come that may be specific to that. And then that goes through this assessment. And then as I said, providing that residual risk score, as well as a mitigation or maturity score, so that we can really reward good behavior. But it might recognize that there's more maturity that can be made within each of those dimensions that are being evaluated. So you can see on the right hand screen, just a snapshot of what that looks like, again, trying to make it as easy as possible for practitioners. So that it's really helping to guide them along the design process, and set them up for success with future regulations and certifications that we're working on formalizing.

**37:08** And so through that, just to be really clear, that we're really trying to take these principles, put them into meaningful practice is, what we've done is go from those overall guiding principles, in this circumstance, it's fairness, as just the example. And then within that, showing how we can, how those requirements that you saw on the previous slide, really matched to trying to achieve that fairness. And so that is, the requirements being identified the unwanted bias, the mechanism to do so that you can see in the guideline is then to perform a harms mapping within that 2.2 is then to test for unwanted bias. And so we've seen a lot of this, in terms of the ability to apply statistical tests to understand if there's bias between certain types of demographics. So demographic parity is the test that is performed there.

**38:15** And so within that, what's important to know is that, even though these types of tests can be performed, achieving fairness can't just be achieved through or that can't just be done through doing a test and doing a point in time test at that. It's really important that we're thinking about bias in terms of identifying it, identifying all the ways in which it can be mitigated, as well. And that includes things like training, as you can see here, but then also recourse mechanisms, if there is an issue that does arise, recognizing that this system wasn't necessarily fair for all demographics so some demographics might need to be supported over others through either human intervention, or a recourse mechanism that would make sense to that type of system. And so what we're trying to get at here is then the ability to actually have a standard or control in place that is able to be audited against, and so and then those would be able to or those that are being evaluated would be able to provide evidence, just like a normal audit, think of like a financial audit. So through that, that's where we're really trying to work with the community. And then as I said, best in class research to really identify then how do you apply those statistical tests as just one component of evaluating fairness? And which test applies in that circumstance?

**39:56** And so this is, this slide really matches with the other slide of going from these higher order principles and then all the way to that evaluation mechanism. But just giving you an example of how that can actually be done. So all this to say is you can imagine that all of this work is really, requires a lot of research. It requires a lot of



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community discussions with various different types of subject matter experts. Through that we've recognized the need to really support organizations to build out policy and governance.

**40:34** Our main goal is to develop a formalized certification, which for responsible use of AI, which we are in the process of with various different accreditation bodies in Canada, the US and the UK. But within this, we've really started to develop quite a few tools that can help practitioners. So we'd encourage you to, to use those resources that are available online, but please reach out if you do have any questions. And with that, thank you for participating in this module. And I we're at [responsible.ai](https://responsible.ai) if you'd like to find us and find out more about the work that we do.